

SECTION 15090-A

PVC AND CPVC PIPING SYSTEMS

PART 1 - GENERAL

1.1 WORK INCLUDED

- A. This Section specifies the requirements for materials, fabrication, and installation of PVC and CPVC piping systems.

1.2 RELATED WORK

- A. This Section shall be used in conjunction with the following other specifications and related Contract Documents to establish the total requirements for PVC and CPVC piping systems.
- B. CAUTION: Use of this Section without including all of the above-listed items will result in omission of basic requirements.
- C. In the event of conflict regarding requirements for PVC and CPVC piping systems between this Section and any other section, the provisions of this Section shall govern.

1.3 SYSTEM DESCRIPTION

A. Definitions:

- 1. Carrier pipe or tubing: refers to the inner or primary piping or tubing continuously subjected to fluid.
 - 2. Containment pipe and fittings: refers to the outer pipe or secondary pipe subject to the fluid only on leakage from the carrier pipe.
 - 3. Centralizers: refers to the clip device which keeps the carrier pipe or tubing located in the center of the containment pipe.
 - 4. Containment boxes: refers to an enclosure on a contained piping system that maintains containment while accommodating another function such as valves containment or level switch/leak detection containment.
- B. Secondary containment shall be provided as shown in the Drawings.

PART 2 - PRODUCTS

2.1 GENERAL

- A. Material requirements are specified in the pipe class specification sheets.
- B. Pipe, fittings, valves, and materials shall be stored in an indoor location with an ambient temperature of between 40 and 90 degrees F. Pipe, fittings, and valves shall be stored and handled in such a manner as to prevent damage and/or contamination. Materials that become damaged must be replaced with new, clean material.

2.2 MATERIALS

- A. Primer and solvent cement shall be obtained from the same manufacturer.
- B. Solvent cement manufacturer and viscosity shall be as recommended by the pipe and fitting manufacturer to ensure compatibility and shall comply with Part X3 of ASTM D2855.
- C. In the event the following procedures vary from those of the materials manufacturer, the manufacturer's procedure shall govern. The Engineer shall be notified in writing of any such variance.

2.3 CONTAINMENT

- A. Identify the minimum size for containment piping. The table below is for reference, and should be modified based on the tubing/piping pull length and the number of bends. In general, the carrier pipe cross-sectional area should not exceed 25 percent of the containment pipe cross-sectional area.

Carrier Pipe or Tube(s), Nominal Size	Containment Pipe, Minimum Nominal Size
Tubing:	
1 tube at 1/4-inch	1-inch
1 tube at 3/8-inch	1-inch
1 tube at 1/2-inch	1-inch
1 tube at 3/4-inch	1-1/2-inch
1 tube at 1-inch	2-inch

Carrier Pipe or Tube(s), Nominal Size	Containment Pipe, Minimum Nominal Size
2 tubes at 1/4-inch	1-1/2-inch
2 tubes at 3/8-inch	2-inch
2 tubes at 1/2-inch	3-inch
2 tubes at 3/4-inch	3-inch
2 tubes at 1-inch	4-inch
1 tube at 3/8-inch and 1 at 1/4-inch	1-1/2-inch
1 tube at 3/8-inch and 1 at 1/2-inch	2-inch
1 tube at 3/8-inch and 1 at 3/4-inch	3-inch
1 tube at 3/8-inch and 1 at 1-inch	3-inch
1 tube at 1/4-inch and 1 at 1/2-inch	2-inch
1 tube at 1/4-inch and 1 at 3/4-inch	2-inch
1 tube at 1/4-inch and 1 at 1-inch	2-inch
1 tube at 1/2-inch and 1 at 3/4-inch	3-inch
1 tube at 1/2-inch and 1 at 1-inch	3-inch
1 tube at 3/4-inch and 1 at 1-inch	3-inch
1 tube at 3/8-inch and 2 at 3/4-inch	3-inch
1 tube at 3/8-inch, 1 at 3/4-inch, and 1 at 1-inch	3-inch
1 tube at 3/8-inch and 2 at 1-inch	3-inch
1 tube at 1/2-inch and 2 at 3/4-inch	3-inch
1 tube at 1/2-inch, 1 at 3/4-inch, and 1 at 1-inch	4-inch
1 tube at 1/2-inch and 2 at 1-inch	4-inch
Pipe:	
3/4 inch (20A)	2 inch (50A)
1 inch (25A)	2 inch (50A)
1 ¼ inch (32A)	3 inch (80A)
1 ½ inch (40A)	3 inch (80A)
2 inch (50A)	4 inch (100A)
2 ½ inch (65A)	6 inch (150A)
3 inch (80A)	6 inch (150A)
4 inch (100A)	8 inch (200A)

Carrier Pipe or Tube(s), Nominal Size	Containment Pipe, Minimum Nominal Size
6 inch (150A)	10 inch (250A)

- B. Provide valve or in-line instrument containment by joining the containment piping to a containment box constructed of PVC or polypropylene. The box shall extend a minimum of 4 inches beyond the valve or instrument in all directions and a drain valve shall be fitted at the bottom of each box.

PART 3 - EXECUTION

3.1 GENERAL

- A. Rigid PVC pipe shall be cut, made up, and installed in a covered location in accordance with the pipe manufacturer's recommendations and with ASTM D2855.
- B. Schedule 40 pipe shall not be threaded. Use Schedule 80 threaded nipples where necessary to connect threaded valves or fittings.
- C. Only strap wrenches shall be used for tightening threaded plastic joints, and care shall be taken not to over tighten these fittings. As a guide, it is recommended to tighten two full turns after resistance is first encountered turning by hand.
- D. Provide adequate ventilation when working with pipe joint solvent cement.
- E. 2-Inch and Larger PVC Gluing Procedures:
1. Install a positive room ventilation system approved by the Contractor.
 2. Leave fabrication low points disconnected to allow fumes to dissipate.
 3. Place a fan near fabrication low points to dissipate fumes in enclosed rooms; provide and maintain full room exhaust and monitor air on a regular basis to ensure that all quality standards are being maintained.
 4. Clearly mark a safety exclusion control area 10 feet around the open ends of pipe.
 5. Leave open ends of pipe disconnected for 8 hours, allowing fumes to dissipate (with fan running).
 6. All valves in a system are to be left in the open position until ventilation of the system is completed.

7. Connect end of pipe, completing fabrication, at the low point.
- F. PVC pipe installed in direct sunlight outdoors shall be painted with two coats of white high-pigment water-based latex paint per requirements of the contract to prevent UV degradation.
- G. Inspect pipe and fittings and do not use materials showing defects or irregularities.
- H. Cut, prime, cement, and join the pipe and fittings following manufacturer's procedures and ASTM D2855.
- I. For pipe size 6 inches and larger, a crew of two men is recommended, and the following additional steps are necessary:
 1. Hold joint together for 1 to 3 minutes depending on pipe size.
 2. As an aid in joining these larger sizes, it is recommended that a come-along be used in pulling the pipe and fitting together.
- J. Do not thread male metal pipe threads into female plastic fittings.

3.2 FABRICATION

- A. The cementing and joining shall not be done when the ambient temperature is below 40 degrees F or above 90 degrees F, and piping shall not be exposed to direct sunlight either in storage or after installation.
- B. Solvent welding shall be done in a clean, covered area.
- C. If atmospheric temperature is above 90 degrees F, use the following precautions to avoid excessive evaporation of solvent from the cement:
 1. Shade joint surfaces from sun at least 1 hour prior to joining.
 2. Make joints in early morning hours and as quickly as possible after applying cement.
- D. If ambient temperature is below 40 degrees F, joints must be prefabricated in a heated space, or a temporary heat enclosure shall be provided during the entire joining period and for 2 hours after joint is made up. Materials shall be stored in a heated area.

3.3 HANDLING

- A. Joints and adjacent piping shall not be moved after joining for the set time periods identified in ASTM D2855.
- B. Test pressure shall not be applied after joining for the minimum times specified in Table X2.1 of ASTM D2855.

3.4 SUPPORTS AND HANGERS

- A. Hanger rod sizing shall be as specified in Section 15140, Piping Supports and Anchors.
- B. Spacing for PVC piping system supports shall not exceed the values in Table 3.

Table 3
Maximum Spacing Between Supports for PVC Pipe
(Feet; Based on Pipe Full of Liquid with Specific Gravity of 1.0)

Schedule 40

Size	Operating Temperature (Degrees F/C)				
	60/15	80/25	100/40	120/50	140/60
1 inch (25A)	5.5	5.0	4.5	3.0	2.5
1 ½ inch (40A)	6.0	5.5	5.0	3.5	3.0
2 inch (50A)	6.0	5.5	5.0	3.5	3.0
3 inch (80A)	7.0	7.0	6.0	4.0	3.5
4 inch (100A)	7.5	7.0	6.5	4.5	4.0
6 inch (150A)	8.5	8.0	7.5	5.0	4.5
8 inch (200A)	9.0	8.5	8.0	5.0	4.5
10 inch (250A)	10.0	9.0	8.5	5.0	5.0
12 inch (300A)	11.5	10.5	9.5	6.5	5.5
14 inch (350A)	12.0	11.0	10.0	7.0	6.0

Schedule 80

Size	Operating Temperature (Degrees F/C)					
	60/15	80/25	100/40	120/50	140/60	160/70
1 inch (25A)	6.0	5.5	3.5	3.0	3.0	2.5
1 ½ inch (40A)	6.5	6.5	5.5	3.5	3.5	3.0
2 inch (50A)	7.0	6.5	6.0	4.0	3.5	3.0
3 inch (80A)	8.0	7.5	7.0	4.5	4.0	4.0
4 inch (100A)	9.0	8.5	7.5	5.0	4.5	4.0
6 inch (150A)	10.0	9.5	9.0	6.0	5.0	4.5
8 inch (200A)	11.0	10.5	9.5	6.5	5.5	5.0
10 inch (250A)	11.5	11.0	10.5	9.5	8.0	--
12 inch (300A)	12.5	12.0	11.5	10.5	8.5	--
14 inch (350A)	13.5	13.0	11.0	8.0	7.0	--

Specific Gravity Correction Factor

Specific Gravity						
1.0	1.10	1.20	1.40	1.60	2.00	2.50
1.0	0.98	0.96	0.93	0.90	0.85	0.80
Correction Factor						

- C. Spacing for CPVC piping system supports shall not exceed the values given in Table 4.

Table 4

Maximum Spacing Between Supports for CPVC Pipe
(Feet; Based on Pipe Full of Liquid with Specific Gravity of 1.0)

Schedule 80

Size	Operating Temperature (Degrees C)						
(Inches)	80/25	100/40	120/50	140/60	160/70	180/80	200/90
½ inch (15A)	5.00	4.75	4.50	4.25	3.66	3.25	2.50
1 inch (25A)	5.75	5.50	5.25	4.66	4.33	3.50	2.66

Size	Operating Temperature (Degrees C)						
1 ½ inch (40A)	6.50	6.25	5.75	5.25	4.75	4.00	3.25
2 inch (50A)	7.25	6.75	6.33	5.67	5.25	4.25	3.33
3 inch (80A)	8.50	8.00	7.50	6.75	5.67	5.00	4.00
4 inch (100A)	9.25	8.75	8.25	7.33	6.50	5.33	4.33
6 inch (150A)	10.00	9.33	8.75	7.67	7.00	6.00	4.75
8 inch (200A)	10.50	10.00	9.33	8.50	7.50	6.50	5.25

Specific Gravity Correction Factor

Specific Gravity						
1.0	1.10	1.20	1.40	1.60	2.00	2.50
1.0	0.98	0.96	0.93	0.90	0.85	0.80
Correction Factor						

- D. The data in Tables 3 and 4 is for un-insulated lines. For insulated lines, reduce spans to 70 percent of the tabulated value. Use continuous support for spans of less than 2.5 feet. Cable tray with cross supports at least every 12 inches meets the intent of continuous supports.

3.5 ASSEMBLY AND INSTALLATION

- A. Piping shall be aligned vertically and horizontally and at right angles unless otherwise specified by design.
- B. Corrections in alignment shall not be made by applying force to piping.
- C. Flanges shall be bolted using a calibrated torque wrench, following the manufacturer's procedures, and following requirements of the contract.
- D. Install containment piping according to the following sequence:
 - 1. Carrier piping with the appropriate centralizers in place. Do not use centralizers for flexible tubing carrier piping.

2. Containment piping over the carrier pipe and centralizers.
3. Fittings on the carrier piping.
4. Testing of the carrier piping.
5. Fittings on the containment piping (when split fittings specified in attached Appendix).
6. Testing of the containment piping by either of the following approaches:
 - a. Use 5 psig CDA with soaped weld and fitting connections (i.e. using "Snoop").
 - b. Hydrostatic pressure test with the pressure loss not exceeding 0.2 psi over a 15-minute duration. Use a pressure gauge with 0.1-psi resolution increments.
7. For carrier tubing pulled through containment piping, no fittings shall be pulled through the containment piping. Remove any section of carrier tubing which is kinked and install a new section.
8. Slope horizontal runs of containment pipe at 1-percent minimum slope to low points. Do not pocket containment piping with high or low points unless required by the Drawings.

3.6 LEAK TESTING

- A. Flange bolts shall not be torqued beyond manufacturer specifications to seal joints.
- B. Replace leaking joints. Repair practices such as back welding shall not be used to seal leaking joints.

3.7 CLEAN PVC/CPVC

- A. When required on the piping data sheets, clean PVC/CPVC fabrication and installation practices shall be followed as described below:
 1. Inspection, storage, fabrication, and bagging of ends for PVC/CPVC pipe shall occur in a designated clean PVC construction area. This area shall be maintained clean and free of dust, oils, and other contamination sources. No oil- or fuel-powered engines shall

exhaust into the area and no grinding or welding of pipe other than PVC shall be allowed in this area.

2. Inspect all components for physical deformities or contamination. Reject problem materials.
3. As much as practical, clean PVC/CPVC shall be prefabricated in the clean area to minimize field welding.
4. Fitters shall wear clean gloves when handling clean pipe.
5. Prior to fabrication, pipe and fittings shall be cleaned as follows:
 - a. Using nylon brushes, clean city water, and a surfactant (Triton X100, Micro, or approved equal), scrub pipe and fitting interior and exterior until clean.
 - b. Rinse with clean city water.
 - c. Dry completely with filtered oil-free air (OFA) of quality in conformance with contract requirements.
6. Pipe and fittings shall be cut, faced, deburred, joined, and installed in accordance with the pipe manufacturer's written recommendations. When chamfering or deburring pipe ends, install a Kleenwipe and plug the pipe inside the location of the deburring such that, at completion of pipe end dressing, removal of the Kleenwipe removes debris. Wipe interior and pipe edge after deburring to remove particulate.
7. Before transporting clean spools out of the clean area, ends and openings shall be bagged and taped.

3.8 ENHANCED QUALITY ASSURANCE

- A. When required on the piping data sheets, enhanced quality assurance shall be applied.
- B. Only materials free from visible cracks, holes, foreign inclusions, and surface imperfections, which are of uniform color and homogeneous throughout, and are of proper dimensions, roundness, and alignment, shall be used.
- C. Only fitters (or bonders) qualified according to ANSI B31.3 shall work on assembling and installing pipe.

Qualification examinations shall include internal examination for full bonding at test joints. Fitters shall be qualified to install joints up to two times the largest test joint successfully completed in the qualifying examination.

- D. The Contractor shall be responsible for training, qualification, and documentation for all fitters.
- E. Fabrication, assembly, and erection of piping systems shall be according to ANSI B31.3.
- F. Fitters' identification shall be affixed to each joint prepared.
- G. A written record shall be maintained by fitters for each joint and assembly completed. This document shall record joint identification, fitter identification, joint start and completion times, materials inspection record, dry fit record, working conditions, cutting method, pipe surface temperature, primer brand and type, solvent cement brand and type, dimensional irregularities, forces measured (to nearest 0.1 pound) used to correct dimensional irregularities and misalignments, start time for setting joint, time joint first moved after setting, time joint is pressure tested, and flange bolt torque values.
- H. Assembly dimension irregularities and misalignment shall be reported to Engineer. Forces used to correct misalignment must be measured and approved by Engineer.
- I. Flange parts shall be discarded and replaced with a new flange when a bolt is torqued beyond specification or when flange connection is not in compliance with the manufacturer's torquing procedure.

PVC AND CPVC PIPING SYSTEMS

Pipe Class	Description
CC01	Schedule 80 CPVC
CC02	Schedule 80 CPVC, enhanced QA
PC01	Schedule 40 PVC
PC02	Schedule 40 PVCDWV fittings
PC03	Schedule 40 clear PVC, Schedule 40 fittings
PC04	Schedule 40 clear PVC, split fittings
PC05	Schedule 80 PVC
PC06	Schedule 80 PVC, enhanced QA
PC07	Schedule 80 PVC, clean

PIPE CLASS SPECIFICATION SHEET

Pipe class	CC01
Rating	100 psig at 150 degree F (65 degrees C) through 12 inch (300A)
Pipe	Schedule 80 CPVC, ASTM F441, Class 23447B
Fitting	Schedule 80, ASTM F439, Class 23447B; socket-weld all dimensions
Flanges	Dimensions conform to ANSI B16.5 for steel flanges
Unions	Through 2 inch (50A), socket-weld, Schedule 80, O-ring material to match flange gaskets.
Cement	ASTM F493, equivalent to Weld-On 714
Primer	Equivalent to Weld-On P70 purple
Bolt thread lubricant	None

PIPE CLASS SPECIFICATION SHEET

Pipe class	CC02
Rating	100 psig at 150 degree F (65 degrees C) through 12 inch (300A)
Pipe	Schedule 80 CPVC, ASTM F441, Class 23447B
Fitting	Schedule 80, ASTM F439, Class 23447B; socket-weld all dimensions
Flanges	Dimensions conform to ANSI B16.5 for steel flanges
Unions	Through 2 inch (50A), socket-weld, Schedule 80, O-ring material to match flange gaskets.
Cement	ASTM F493, equivalent to Weld-On 714
Primer	Equivalent to Weld-On P70 purple
Bolt thread lubricant	None
Quality assurance	Enhanced

PIPE CLASS SPECIFICATION SHEET

Pipe class	PC01
Rating	100 psig at 73 degree F (23 degrees C) through 12 inch (300A)
Pipe	Schedule 40 PVC, ASTM D1785, Class 12454B
Fitting	Schedule 40, ASTM D2466, Class 12454B; socket-weld
Flanges	Dimensions conform to ANSI B16.5 for steel flanges
Unions	Up through 2 inch (50A) diameter, socket-weld, Schedule 40, O-ring material to match flange gaskets.
Cement	ASTM D2564, equivalent to Weld-On 705 under 6 inches, and Weld-On 711 for 6 inches and over
Primer	Equivalent to Weld-On P70 purple
Bolt thread lubricant	None

PIPE CLASS SPECIFICATION SHEET

Pipe class	PC02
Rating	Atmospheric
Pipe	Schedule 40 PVC, ASTM D1785, Class 12454B
Fitting	Schedule 40, ASTM D2665, Class 12454B; socket-weld, DWV type
Flanges	Dimensions conform to ANSI B16.5 for steel flanges
Unions	Up through 2 inch (50A) diameter, socket-weld, Schedule 40, O-ring material to match flange gaskets.
Cement	ASTM D2564, equivalent to Weld-On 705 under 6 inches, and Weld-On 711 for 6 inches and over
Primer	Equivalent to Weld-On P70 purple
Bolt thread lubricant	None

PIPE CLASS SPECIFICATION SHEET

Pipe class	PC03																
Rating	90 psig at 73 degree F (23 degrees C) through 4 inch (100A)																
Pipe	Schedule 40 clear PVC, ASTM D1785																
Fitting	Schedule 40 PVC (white), ASTM D2466, Class 12454B, socket-weld or hot-air formed long radius bends as follows (use fittings wherever possible):																
	<table> <tr> <th>Containment Pipe Diameter</th><th>Minimum Bend Radius (Inches)</th></tr> <tr> <td>¾ inch (20A)</td><td>9</td></tr> <tr> <td>1 inch (25A)</td><td>12</td></tr> <tr> <td>1 ½ inch (40A)</td><td>12</td></tr> <tr> <td>2 inch (50A)</td><td>18</td></tr> <tr> <td>3 inch (80A)</td><td>24</td></tr> <tr> <td>4 inch (100A)*</td><td>36</td></tr> <tr> <td>6 inch (150A)*</td><td>36</td></tr> </table>	Containment Pipe Diameter	Minimum Bend Radius (Inches)	¾ inch (20A)	9	1 inch (25A)	12	1 ½ inch (40A)	12	2 inch (50A)	18	3 inch (80A)	24	4 inch (100A)*	36	6 inch (150A)*	36
Containment Pipe Diameter	Minimum Bend Radius (Inches)																
¾ inch (20A)	9																
1 inch (25A)	12																
1 ½ inch (40A)	12																
2 inch (50A)	18																
3 inch (80A)	24																
4 inch (100A)*	36																
6 inch (150A)*	36																
	* 4 inch (100A)- and 6 inch (150A)-inch containment pipes shall not be hot-air bent in the field, but may be purchased pre-bent from a containment system vendor.																
Flanges	Dimensions conform to ANSI B16.5 for steel flanges																
Unions	Up through 2-inch diameter, socket-weld, Schedule 40, O-ring material to match flange gaskets.																
Cement	Clear, equivalent to Weld-On 717																
Primer	Clear, Type IPS P-70C, NSF-approved for PVC by Industrial Polychemical Service; use of purple primer prohibited																
Bolt thread lubricant	None																

PIPE CLASS SPECIFICATION SHEET

Pipe class	PC04
Rating	Atmospheric
Pipe	Schedule 40 clear PVC, ASTM D1785
Fitting	Injection-molded two-piece PVC, split along axis parallel to the adjoining pipe; epoxy/PVC cement sealed or mechanical means with elastomeric seal; R&G Sloane Contain-It or Flo-Safe
Flanges	Dimensions conform to ANSI B16.5 for steel flanges
Unions	Up through 2 inch (50A) diameter, socket-weld, Schedule 40, O-ring material to match flange gaskets.
Cement	Clear; equivalent to Weld-On 717
Primer	Clear, Type IPS P-70C, NSF-approved for PVC by Industrial Polychemical Services; use of purple primer prohibited
Bolt thread lubricant	None

PIPE CLASS SPECIFICATION SHEET

Pipe class	PC05
Rating	100 psig at 100 degree F (38 degrees C) through 12 inch (300A)
Pipe	Schedule 80 PVC, ASTM D1785, Class 12454B
Fitting	Schedule 80, D2467, Class 12454B; socket-weld all dimensions
Flanges	Dimensions conform to ANSI B16.5 for steel flanges
Unions	Up through 2 inch (50A), socket- weld, Schedule 80, O-ring material to match flange gaskets.
Cement	ASTM D2564, equivalent to Weld-On 711 or 717 gray for <8 inches, and Weld-On 719 for ≥8 inches
Primer	Equivalent to Weld-On P70 purple
Bolt thread lubricant	None

PIPE CLASS SPECIFICATION SHEET

Pipe class	PC06
Rating	100 psig at 100 degree F (38 degrees C) through 12 inch (300A)
Pipe	Schedule 80 PVC, ASTM D1785, Class 12454B
Fitting	Schedule 80, ASTM D2467, Class 12454B; socket-weld all dimensions
Flanges	Dimensions conform to ANSI B16.5 for steel flanges
Unions	Up through 2 inch (50A), socket- weld, Schedule 80, O-ring material to match flange gaskets.
Cement	ASTM D2564, equivalent to Weld-On 711 or 717 gray under 8 inches, and Weld-On 719 for 8 inches and over
Primer	Equivalent to Weld-On P70 purple
Bolt thread lubricant	None
Quality assurance	Enhanced

PIPE CLASS SPECIFICATION SHEET

Pipe class	PC07
Rating	100 psig at 100 degree F (38 degrees C) through 12 inch (300A)
Pipe	Schedule 80 Clean PVC, ASTM D1785, Class 12454B
Fitting	Schedule 80, ASTM D2467, Class 12454B; socket-weld all dimensions
Flanges	Dimensions conform to ANSI B16.5 for steel flanges
Unions	Up through 2 inch (50A), socket weld, Schedule 80, O-ring material to match flange gaskets.
Cement	ASTM D2564, equivalent to Weld-On 711 or 717 gray under 8 inches, and Weld-On 719 for 8 inches and over
Primer	Equivalent to Weld-On P70 purple
Bolt thread lubricant	None
Cleaning	Clean, per this Section

- END -